

"ROUGH SHEET-02"

Course No: Physics 1213

Course Title: Physics

Experiment No : 02

Experiment Name : Determination of the refractive index of the material of a prism by a spectrometer

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Section: C

Group: 06

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Experiment Name: Determination of the refractive index of the material of a prism by a spectrometer

Theory:

Refractive index is the ratio of the speed of light in vacuum to the speed of light in a medium. Let, a ray PQ be the incident on the first face AB of a prism and then after passing through the prism, finally emerges out through the other face AC in the direction RS. Let i_1 and r_1 be the respective angles of incidence and refraction at the first face of the prism and r_2 and i_2 be the corresponding quantities for the second face. Now the deviation of the ray is given by the angle $\angle SOT$,

$$\delta = (i_1 - r_1) + (i_2 - r_2) \dots (1)$$

But in the minimum deviation position, the ray passes symmetrically through the prism so that $i_1 = i_2$ and $r_1 = r_2$. Therefore, the angle of the minimum deviation,

$$\delta_m = 2(i_1 - r_1) \dots (2)$$

But,

$$\angle LMR = r_1 + r_2 = 2r_1 = A \dots (3)$$

i.e.

$$r_1 = A/2$$

where A = angle of prism

From eq (2) and (3), we get $i_1 = \frac{A + \delta_m}{2}$

Hence,
$$\mu = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}}$$

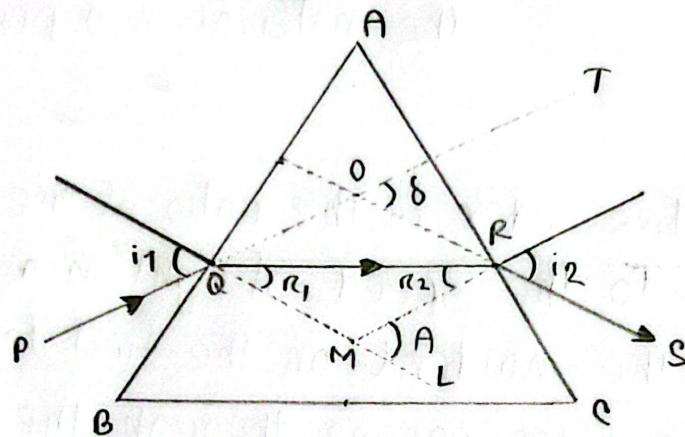


Fig 1: Refraction mechanism through the prism

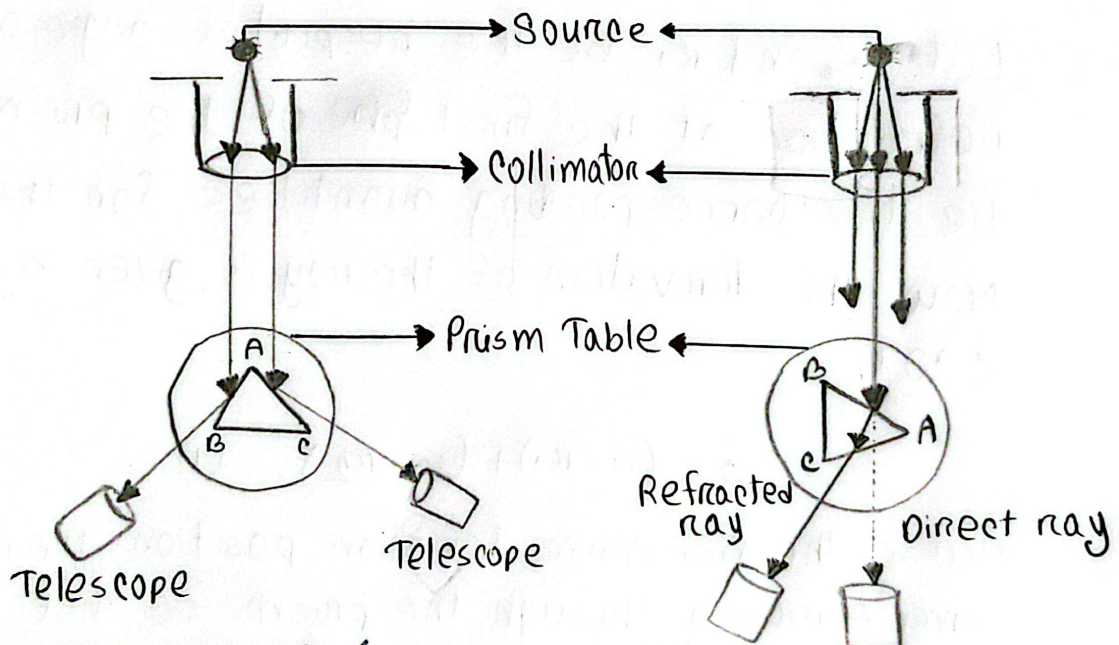


Fig 2: Measurement of the prism angle, A

Fig 3: Measurement of the angle of minimum deviation, δ_m

Required Apparatus:

- Spectrometer
- Prism
- Spirit level
- Na light

calculations:

2000/00/00

Experimental Data:

Table 01 - Data for the angle of prism A

[illegible]

Table 02 - Data for the angle of minimum deviation δ_m

[illegible]

Discussion:

Precautions:

- The source was kept in front of the collimator to make the image bright
- The cross-wire was made coincident with the same edge of the slit image
- Parallax error was avoided
- The telescope was rotated in the same direction to avoid backlash error
- The width of the slit image was made as narrow as possible